

Introduction to Data Science

Session 10: Towards open data science

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Hertie School | GRAD-C11/E1339

Table of contents

1. The replication crisis
2. Reproducible science
3. Towards more trustworthy data science

The replication crisis

The replication crisis

What the crisis is about

- The fact that many scientific findings are difficult or impossible to reproduce.
- Reproducibility is a cornerstone of science as an enterprise of knowledge generation. This is bad.

Factors fueling the replication crisis

- Wrong incentives in science leading to questionable research practices (see next slides)
- Limited data sharing and transparency
- Insufficient methodological training



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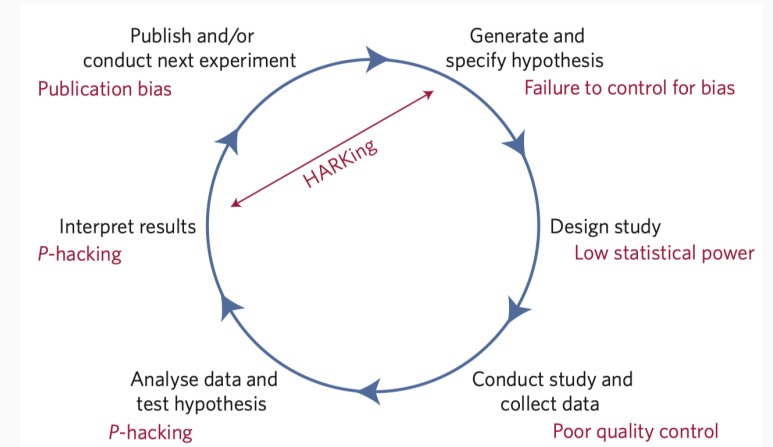


Figure1 | Threats to reproducible science. An idealized version of the hypothetico-deductive model of the scientific method is shown. Various potential threats to this model exist (indicated in red), including lack of replication⁵, hypothesizing after the results are known (HARKing)⁷, poor study design, low statistical power², analytical flexibility⁵¹, P-hacking⁴, publication bias³ and lack of data sharing⁶. Together these will serve to undermine the robustness of published research, and may also impact on the ability of science to self-correct.

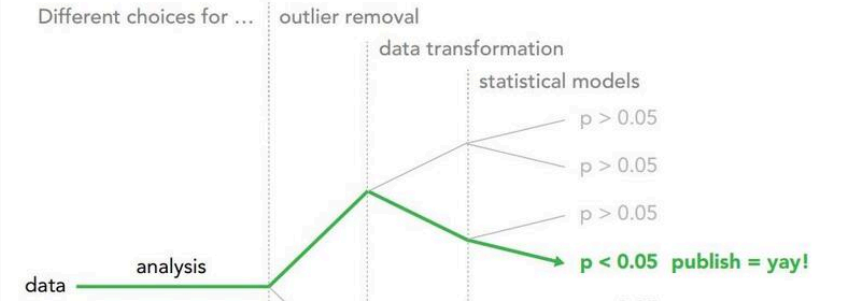
Source **Munafò et al., NHB**

Questionable practices fueling the crisis

Harmful behaviors

- FORKing paths (Researcher Degrees of Freedom): Many plausible analytical choices; selecting those that yield desirable results. (Gelman/Loken 2013)

Garden of forking paths [Gelman and Loken 2014]

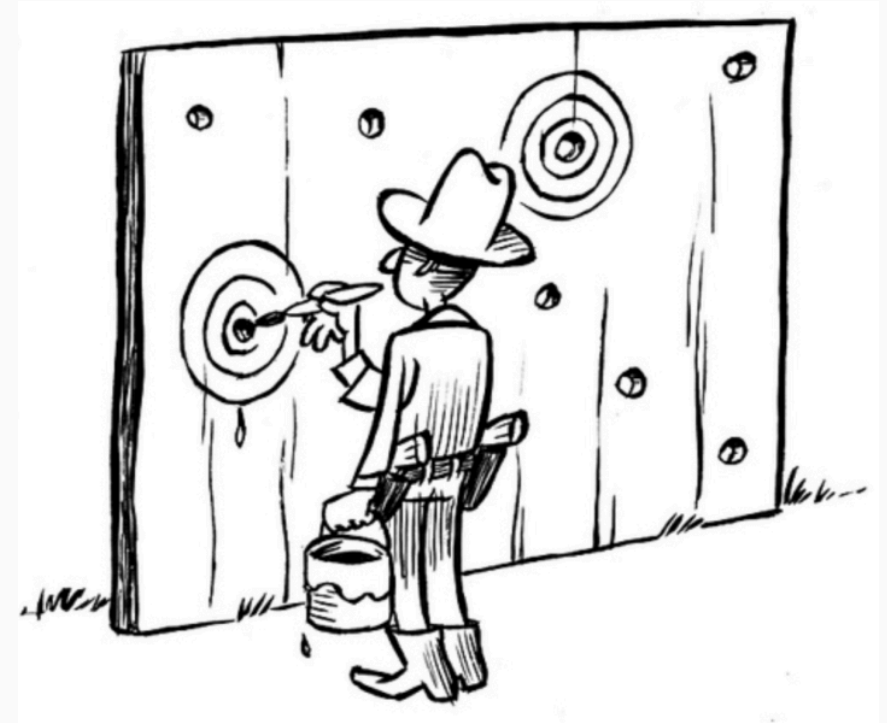


Source Matthew Kay

Questionable practices fueling the crisis

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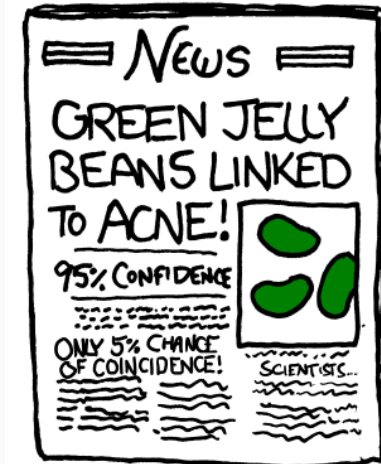
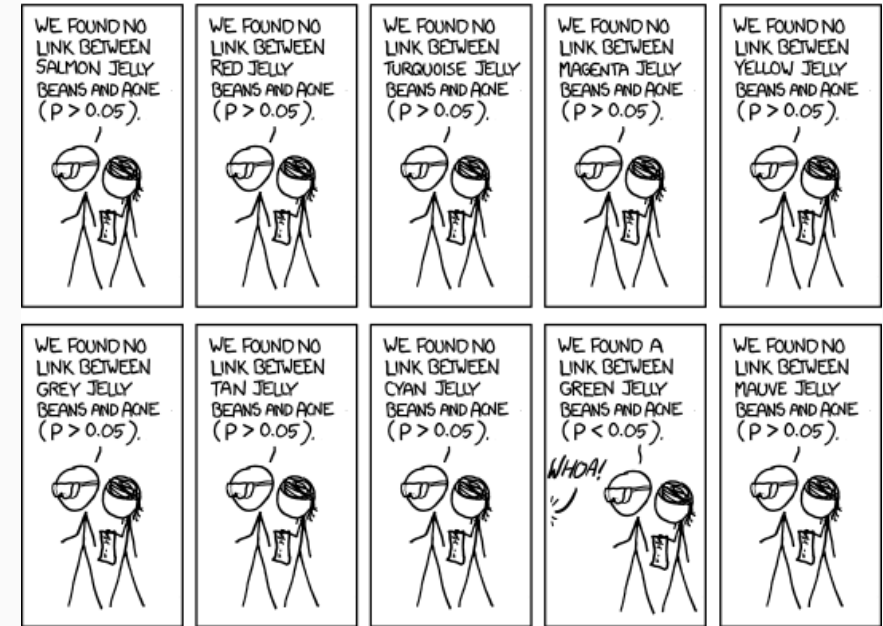


Source Dirk-Jan Hoek

Questionable practices fueling the crisis

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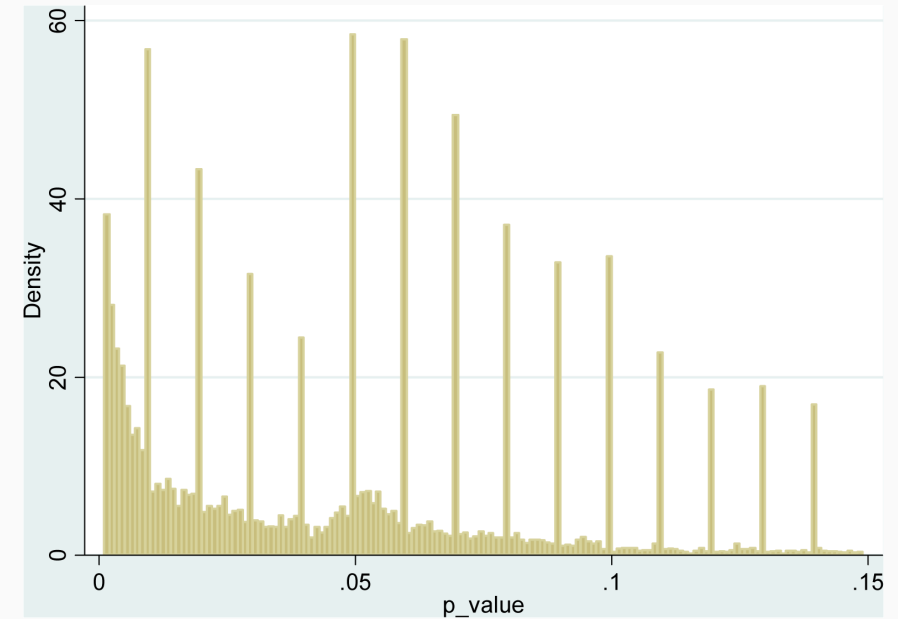
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- **Selective reporting:** Publishing only significant or theoretically convenient outcomes while omitting null or contradictory results.



Source [Krawczyk, M., PLOS One](#)

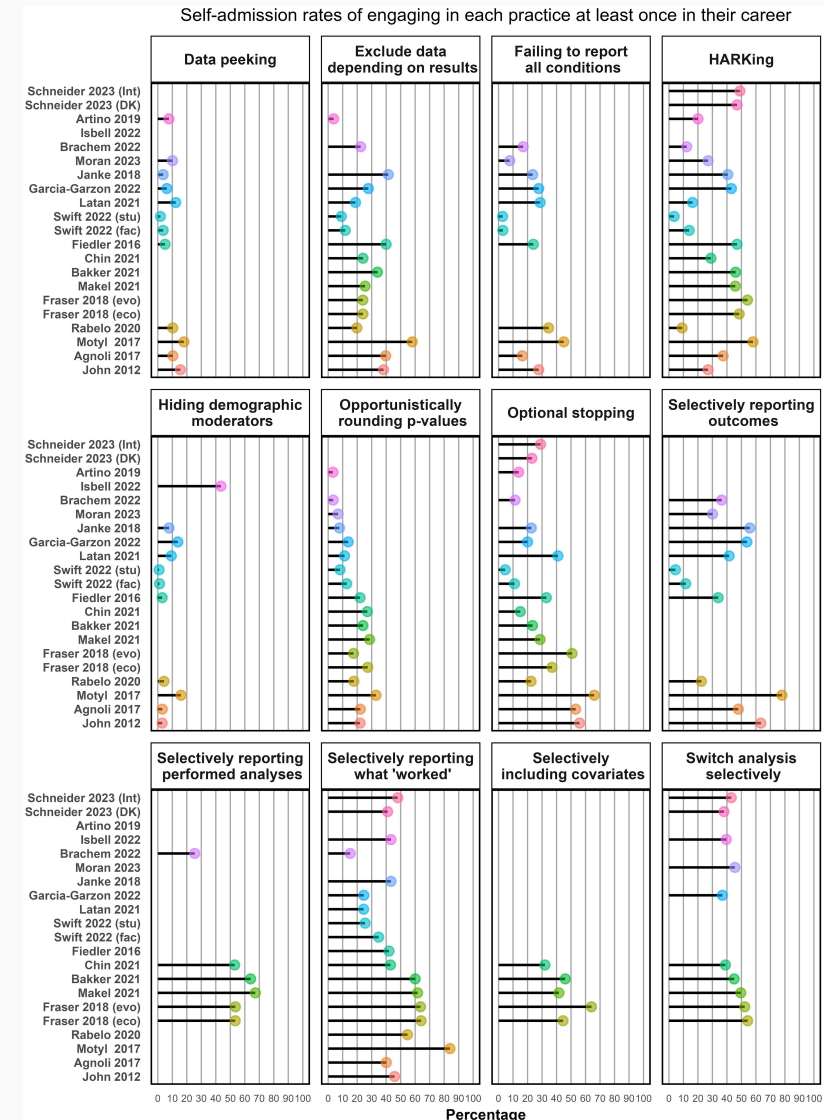
"[i]t appears that some authors choose the mode of reporting their results in an arbitrary way. Moreover, they often end up doing it in such a way that makes their findings seem more statistically significant than they really are (which is well known to improve the chances for publication). Specifically, they frequently report p values 'just above' significance thresholds directly, whereas other values are reported by means of inequalities (e.g. ' $p < .1$ '), they round the p values down more eagerly than up and appear to choose between the significance thresholds and between one- and two-sided tests only after seeing the data. Further, about 9.2% of reported p values are inconsistent with their underlying statistics (e.g. F or t) and it appears that there are 'too many' 'just significant' values."

Questionable practices fueling the crisis

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Source [Lakens et al., Evidence-based Toxicology](#)



Hacking your way to scientific glory

Hack Your Way To Scientific Glory (RECREATION)

You're a social scientist with a hunch: **The U.S. economy is affected by whether Republicans or Democrats are in office.** Try to show that a connection exists, using real data going back to 1948. For your results to be publishable in an academic journal, you'll need to prove that they are "statistically significant" by achieving a low enough p-value.

1 CHOOSE A POLITICAL PARTY

☒ Democrats ☐ Republicans

2 DEFINE TERMS

Which politicians do you want to include?

☒ President ☒ Governors
☐ Senators ☐ Representatives

How do you want to measure economic performance?

☐ Unemployment ☒ Inflation
☒ GDP ☒ Stock prices

Other options

☐ Factor in power

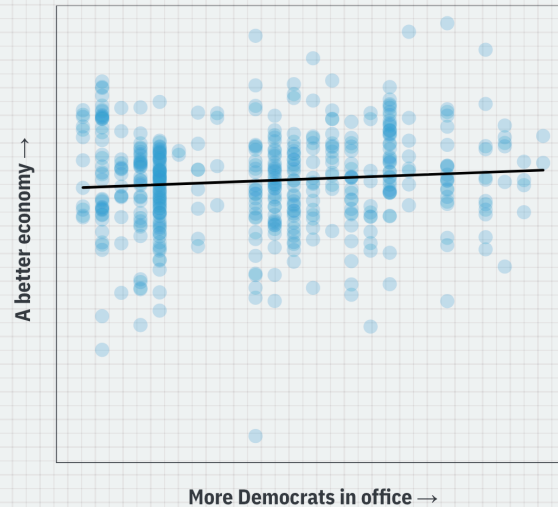
Weight more powerful positions more heavily

☒ Exclude recessions

Don't include economic recessions

3 IS THERE A RELATIONSHIP?

Given how you've defined your terms, does the economy do better, worse or about the same when more Democrats are in power? Each dot below represents one month of data.



Recreation of FiveThirtyEight's "Hack Your Way To Scientific Glory" ([archived here](#)), but with Observable JS.

[Code and data](#)

4 IS YOUR RESULT SIGNIFICANT?

If there were no connection between the economy and politics, what is the probability that you'd get results at least as strong as yours? That probability is your p-value, and by convention, you need a **p-value of 0.05 or less** to get published.



Result: Almost

Your **0.055** p-value is close to the 0.05 threshold. Try tweaking your variables to see if you can push it over the line!

If you're interested in reading real (and more rigorous) studies on the connection between politics and the economy, see the work of [Larry Bartels](#) and [Alan Blinder and Mark Watson](#).

Data from [The @unitedstates Project](#), [National Governors Association](#), [Bureau of Labor Statistics](#), [Federal Reserve Bank of St. Louis](#) and [Yahoo Finance](#).

Reproducible science

A framework for improving reproducibility

Targets for reform (Munafò et al. 2017)

- **Methods.** Better training, collaboration, protecting against cognitive biases
- **Reporting & dissemination.** Promoting pre-registration, improving quality of reporting
- **Reproducibility.** Share data, code, and materials whenever possible
- **Evaluation.** Diversify peer review, preprints, pre-and post-publication
- **Incentives.** Reward openness and reproducibility, not only positive findings

nature
human behaviour

PERSPECTIVE
PUBLISHED: 10 JANUARY 2017 | VOLUME: 1 | ARTICLE NUMBER: 0021

OPEN

A manifesto for reproducible science

Marcus R. Munafò^{1,2*}, Brian A. Nosek^{3,4}, Dorothy V. M. Bishop⁵, Katherine S. Button⁶, Christopher D. Chambers⁷, Nathalie Percie du Sert⁸, Uri Simonsohn⁹, Eric-Jan Wagenmakers¹⁰, Jennifer J. Ware¹¹ and John P. A. Ioannidis^{12,13,14}

Improving the reliability and efficiency of scientific research will increase the credibility of the published scientific literature and accelerate discovery. Here we argue for the adoption of measures to optimize key elements of the scientific process: methods, reporting and dissemination, reproducibility, evaluation and incentives. There is some evidence from both simulations and empirical studies supporting the likely effectiveness of these measures, but their broad adoption by researchers, institutions, funders and journals will require iterative evaluation and improvement. We discuss the goals of these measures, and how they can be implemented, in the hope that this will facilitate action toward improving the transparency, reproducibility and efficiency of scientific research.

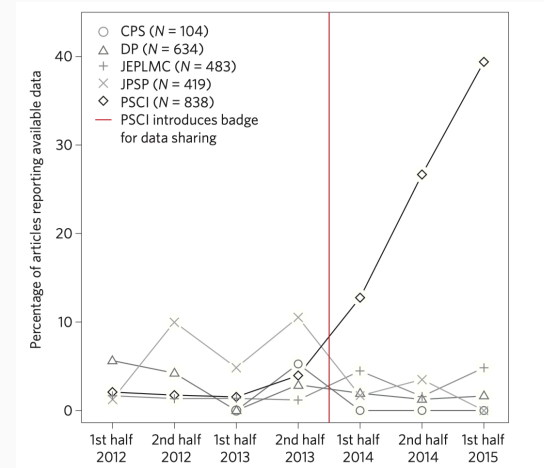


Figure 2 | The impact of introducing badges for data sharing. In January 2014, the journal *Psychological Science* (PSCI) introduced badges for articles with open data. Immediately afterwards, the proportion of articles with open data increased steeply, and by October 2015, 38% of articles in *Psychological Science* had open data. For comparison journals (*Clinical Psychological Science* (CPS), *Developmental Psychology* (DP), *Journal of Experimental Psychology: Learning, Memory and Cognition* (JEPLMC) and *Journal of Personality and Social Psychology* (JPSP)) the proportion of articles with open data remained uniformly low. Figure adapted from ref. 75, PLoS.

Preregistration

Basic idea

- You write down hypotheses and analysis plans in advance.
- You time-stamp and register them in a public or private registry.
- You commit to the primary analysis before seeing the results.
- You clearly separate confirmatory from exploratory work.

Why it potentially helps

- It constrains analytic flexibility and p-hacking.
- It reduces HARKing by fixing hypotheses in advance.
- It clarifies what counts as a decisive test.
- It increases credibility for both positive and null findings.
- Kaplan et al. (right plot): "17 of 30 studies (57%) published prior to 2000 showed a significant benefit of intervention on the primary outcome in comparison to only 2 among the 25 (8%) trials published after 2000."

The preregistration revolution

Brian A. Nosek^{a,b,1}, Charles R. Ebersole^b, Alexander C. DeHaven^a, and David T. Mellor^a

^aCenter for Open Science, Charlottesville, VA 22903; and ^bDepartment of Psychology, University of Virginia, Charlottesville, VA 22904

Edited by Richard M. Shiffrin, Indiana University, Bloomington, IN, and approved August 28, 2017 (received for review June 15, 2017)

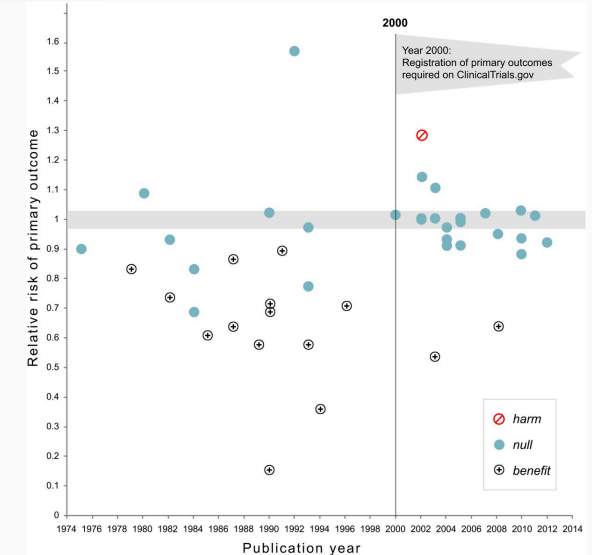
Progress in science relies in part on generating hypotheses with existing observations and testing hypotheses with new observations. This distinction between postdiction and prediction is appreciated conceptually but is not respected in practice. Mistaking generation of postdictions with testing of predictions reduces the credibility of research findings. However, ordinary biases in human reasoning, such as hindsight bias, make it hard to avoid this mistake. An effective solution is to define the research questions and analysis plan before observing the research outcomes—a process called preregistration. Preregistration distinguishes analyses and outcomes that result from predictions from those that result from postdictions. A variety of practical strategies are available to make the best possible use of preregistration in circumstances that fall short of the ideal application, such as when the data are preexisting. Services are now available for preregistration across all disciplines, facilitating a rapid increase in the practice. Widespread adoption of preregistration will increase distinctiveness between hypothesis generation and hypothesis testing and will improve the credibility of research findings.

overconfidence in post hoc explanations (postdictions) and inflate the likelihood of believing that there is evidence for a finding when there is not. Presenting postdictions as predictions can increase the attractiveness and publishability of findings by falsely reducing uncertainty. Ultimately, this decreases reproducibility (6–11).

Mental Constraints on Distinguishing Predictions and Postdictions

It is common for researchers to alternate between postdiction and prediction. Ideas are generated, and observed data modify those ideas. Over time and iteration, researchers develop understanding of the phenomenon under study. That understanding might result in a model, hypothesis, or theory. The dynamism of the research enterprise and limits of human reasoning make it easy to mistake postdiction as prediction. The problem with this is understood as post hoc theorizing or hypothesizing after the results are known (12). It is an example of circular reasoning—generating a hypothesis based on observing data, and then evaluating the validity of the hypothesis based on the same data.

Source Nosek et al., PNAS



Source Kaplan et al., PLOS One

Preregistration developments

Developments (Nosek et al. 2018)

- Rapid growth of preregistered studies across disciplines.
- Emergence of standardized templates and registries.
- Registered reports integrate preregistration into peer review.
- Journals accept papers based on design, not results.

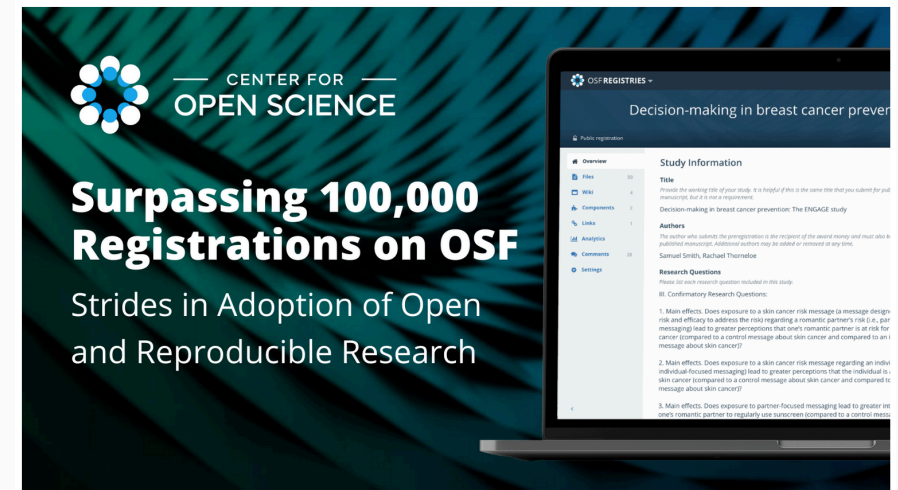
Remaining challenges

- Bad preregistrations can be vague or incomplete.
- Not all projects are suitable for rigid preregistration.
- Distinguishing exploratory work remains important.
- Incentives for high-quality preregistration are still evolving.

Surpassing 100,000 Registrations on OSF: Strides in Adoption of Open and Reproducible Research

September 7th, 2022, Nici Pfeiffer and Mark Call

Posted in: [Preregistration](#), [OSF](#), [Open Science](#), [Registrations](#)



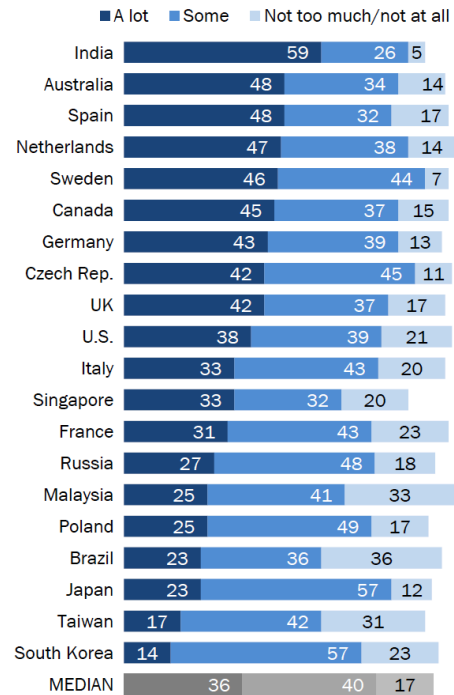
Source [Pfeiffer/Call, COS](#)

Towards more trustworthy data science

Trust in science

Majorities have at least some trust in scientists to do what is right

% who say they have ____ trust in scientists to do what is right for (survey public)

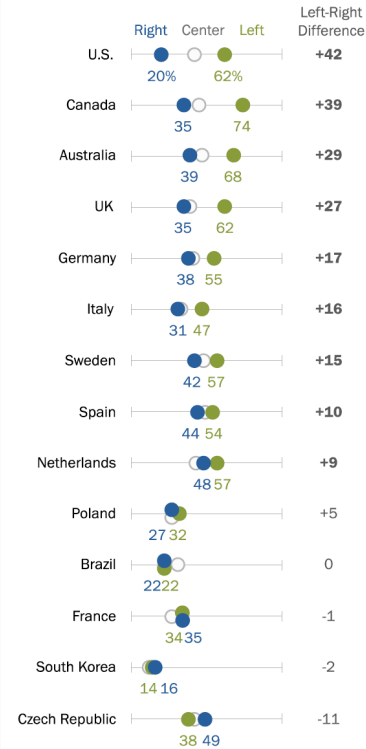


Note: Respondents who did not give an answer are not shown.
Source: International Science Survey 2019-2020. Q2d.
"Science and Scientists Held in High Esteem Across Global Publics"

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Those on the political right often less trusting of scientists than those on left

% who trust scientists **a lot** to do what is right for (survey public)



Note: Statistically significant differences in bold. Respondents who gave other responses or did not give an answer are not shown.
Source: International Science Survey 2019-2020. Q2d.
"Science and Scientists Held in High Esteem Across Global Publics"

PEW RESEARCH CENTER

Trust in science

Relationship of left-right political orientation and trust in scientists

Random slopes of multilevel regression testing the association of trust in scientists and left-right political orientation

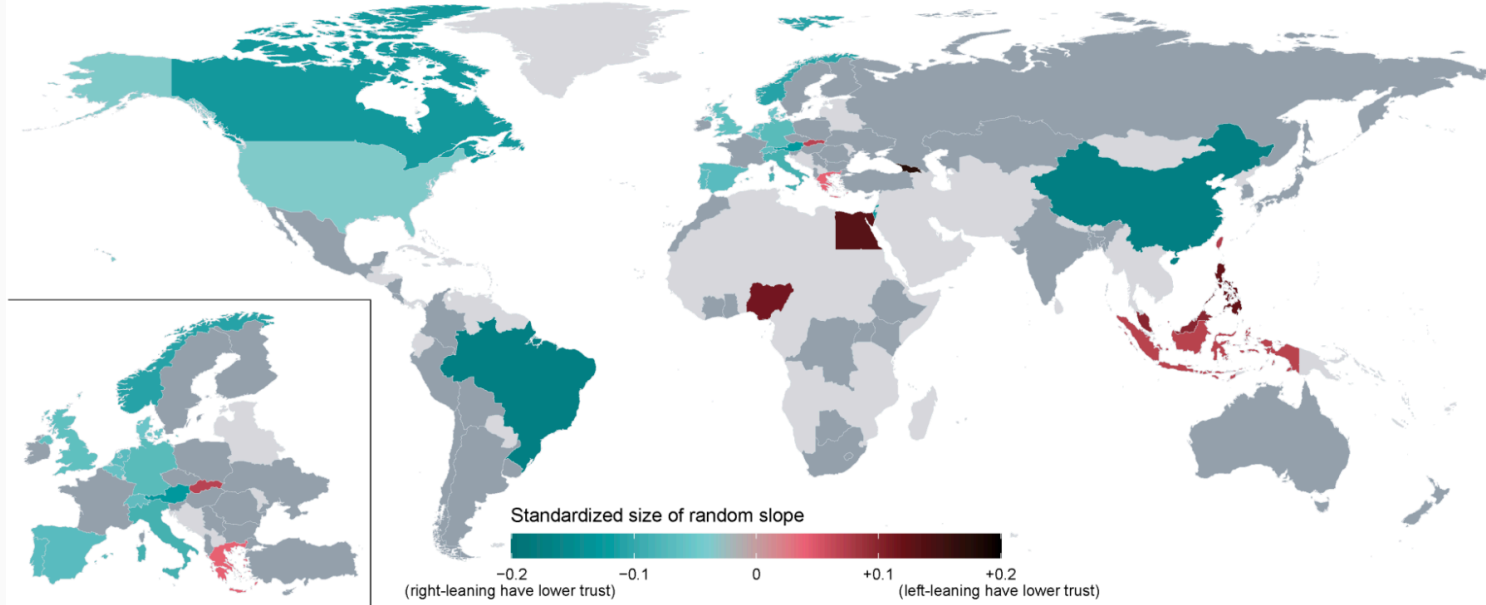


Fig. 3. Relationship of left-right political orientation and trust in scientists. Figure visualises standardised random slopes for political orientation (1 = left – 5 = right), which were extracted from a weighted linear multilevel regression model that explained trust in scientists (1 = very low, 3 = neither high nor low, 5 = very high) across countries and contained random intercepts and slopes of political orientation across countries. Countries with significant effects ($p < .05$) are displayed in colours: Countries coloured in shades of blue show a positive association of left-leaning orientation and trust in scientists (i.e., right-leaning have lower trust). Countries coloured in shades of red show a positive association of right-leaning orientation and trust in scientists (i.e., left-leaning have lower trust) Countries with non-significant effects are shaded in dark grey. Countries with no available data are shaded in light grey.

Trust in data science

Data scientists have the potential to help save the world

By [Leo Borrett](#) May 17, 2017

With an untold number of crises emerging every year, big data is becoming increasingly important for helping aid organisations respond quickly to chaotic and evolving situations.

HOW DATA SCIENCE IS SAVING LIVES

 [AVINASH N](#) Sep 29 · 2 min read



For all the people first priority is about their life. Life is one of the most precious thing in the world. Can Data Science techniques save life, is it possible? Yes, using Data Science techniques to analyze large data sets today has a huge impact on saving lives.

Health

Artificial intelligence and covid-19: Can the machines save us?

Analytics And Data Science

Data Scientist: The Sexiest Job of the 21st Century

Meet the people who can coax treasure out of messy, unstructured data. by Thomas H. Davenport and D.J. Patil

From the Magazine (October 2012)

How AI Will Save Thousands of Lives

Sepsis is the problem; data are the cure

 [Drew Smith, PhD](#) Jan 10, 2020 · 5 min read ★



STUDENTS

Data Science: Why It Matters and How It Can Make You Rich

Trust in data science?

The Cambridge Analytica case: What's a data scientist to do?

The Cambridge Analytica controversy has highlighted data ethics issues especially dear to early career stage data scientists

Researchers just released profile data on 70,000 OkCupid users without permission

By Brian Resnick | @B_resnick | brian@vox.com | May 12, 2016, 6:00pm EDT

An Algorithm That 'Predicts' Criminality Based on a Face Sparks a Furor

Its creators said they could use facial analysis to determine if someone would become a criminal. Critics said the work recalled debunked "race science."

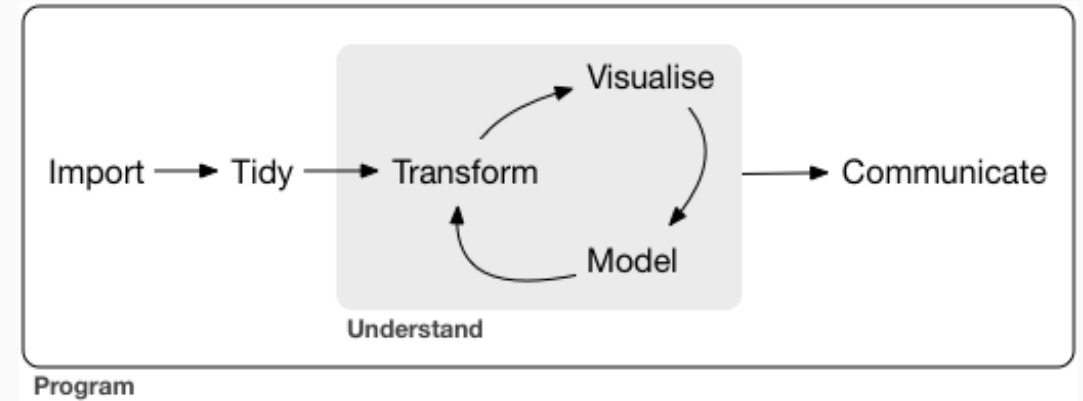
Data Failed the Election, But There's Still Hope for Business

Everyone is blaming data for failing to predict Trump's win. But it's the data handlers who need the real reexamination. [🔗](#)

Fostering trust in science through open science

Communicating the entire workflow

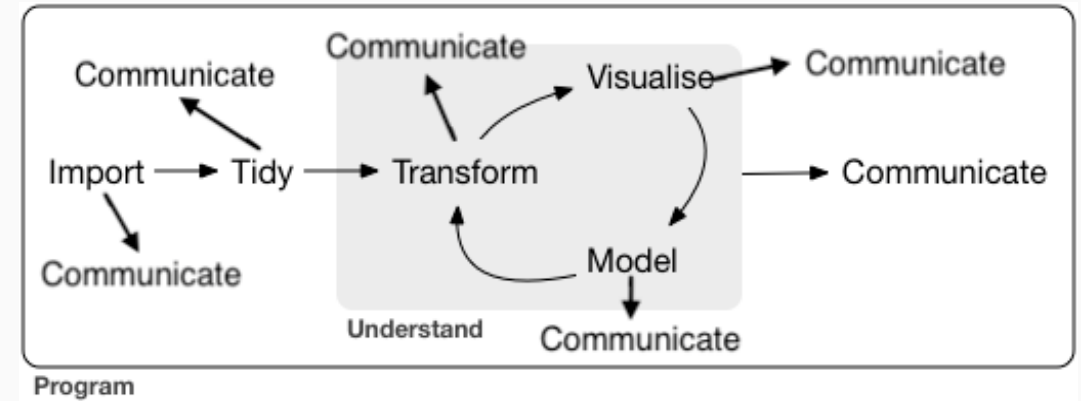
- Any decision - from conceptualizing measures to formatting tables - is meaningful for your output.
 - Tiny mistakes can have massive technical consequences (→ debugging).
 - Various decisions can have ethical implications (→ next session).
 - For others to follow (and potentially invalidate) your analyses, all the details are important.
- What makes the scientific endeavor unique is that it has self-correcting mechanisms.
- But in order for them to work effectively, you have to **be transparent about every step in your workflow.**



Fostering trust in science through open science

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- That's a meta meaning of communication: tell others, by publishing everything from input to code to output, what you've done.



Towards open data science (cont.)

Good practice

- Pre-register designs `osf.io`, `aspredicted.org`
- Do version control `GitHub`
- Publish all research outputs (and inputs if possible)
`GitHub`, `plain-text` formats
- Disclose and document software pipeline `targets`,
`make`
- Make analysis reproducible `renv`, `Docker`
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Notice something?

Over the course, you have been nudged towards become open science practitioners. You are welcome.

